AI-Driven Three-Dimensional Movement Analysis for Objective Assessment of Surgical Expertise in Suturing.

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Abstract

To develop a non-intrusive framework for objective assessment of suturing skill through movement analysis of bilateral hand movements leveraging an artificial intelligence-based markerless tracking to overcome limitations of traditional motion capture systems. Employed directional decomposition of three-dimensional hand movements with machine learning (ML) classification. Six different classification algorithms were implemented to ensure robust performance evaluation. Controlled laboratory environment with standardized suturing simulation setup utilizing an artificial intelligence-powered computer vision tracking system without requiring physical markers or sensors that might interfere with performance. Expert (≥50 patient suturing experiences) and novice participants (<50 patient suturing experiences) were eligible to participate. wEleven experts (4 attending and 7 resident surgeons) demonstrated superior performance compared to novices (11-3rd year medical students) across the majority of movement measures examined. Time-based efficiency emerged as the most distinctive characteristic of surgical expertise, with experts completing 4 interrupted sutures substantially faster and showing reduced hesitation periods during task execution. Spatial control patterns revealed distinct expertise signatures that varied by hand (left and right). Experts exhibited more precise movement control across multiple movement directions [side-to-side (x-axis), forward-backward (y-axis), up-and-down (z-axis)], demonstrating refined spatial awareness and motor control. Movement acceleration profiles indicated that experts maintained more controlled and deliberate motion patterns, particularly evident in up-and-down movements that require greater three-dimensional spatial coordination. Machine learning classification using 6 established algorithms successfully distinguished experts from novices with high accuracy. Feature importance analysis revealed consistent time-based dominance across both hands, with hesitation time and total completion time consistently ranking as the top 2 predictors for expert classification. Beyond time-based measures, subtle hand-specific patterns emerged: left-hand classification showed secondary importance for side-to-side and forward-backward movement control, while right-hand classification emphasized up-and-down acceleration, velocity, and movement patterns, indicating specialized three-dimensional movement coordination essential for surgical precision. Expert surgeons demonstrated significantly greater hand movement distances compared to novices across all directional axes, coupled with superior motion control characterized by significantly smoother up-and-down acceleration and velocity profiles. Machine learning algorithms demonstrated high accuracy in discriminating between experts and novices.

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